

Does realism of a virtual character influence arousal? Exploratory study with pupil size measurement.

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ABSTRACT

In this poster we discuss exploratory analyses seeking to test whether realism of a virtual character can influence arousal measured by pupil diameter change in response to viewing a virtual character. To do this we conducted a study in which 180 virtual characters were presented to 45 participants in a free-viewing procedure, while their physiological responses were recorded. We manipulated the realism of these characters in terms of their appearance, behavior and intractability. Results suggest that appearance and intractability can influence arousal of the participants.

Index Terms: Virtual Reality (VR)—psychology—pupil—virtual character;

1 INTRODUCTION

Realism is the extent to which a virtual character is perceived as “real” [4]. It can be divided into three categories, depending on how the virtual character looks like (photographic realism), behaves (behavioral realism) or interacts with the user (interactional realism) [6]. In this approach, realism is seen as one of the fundamental aspects of the virtual experience [6], related to e.g. presence, copresence and immersion [1, 6]. Yet, to our knowledge, no one so far explored how realism may influence the physiological arousal of the users. Does the way virtual characters look, behave, and interact influence the arousal of participants? If yes, which parts of realism would be the most influential? This exploratory study sought to test whether realism can influence arousal measured by pupil diameter change in response to viewing a virtual character.

2 PARTICIPANTS

Forty-five participants (11 males) aged 18–46 ($M = 27.1$) were recruited for the experiment. All subjects had normal or corrected to normal visual acuity, normal color vision, lack of neurological diseases, no history of psychiatric disorders, and were not prone to motion sickness. All participants signed informed consent before the study in compliance with local ethics committee guidelines and the Declaration of Helsinki. All participants were paid for their par-

ticipation or received course credit. Four participants were excluded from the study due to failure of the measuring device.

3 MATERIALS

The presented virtual characters featured 18 unique combinations of realism levels: 3 levels of behavioral realism, 3 levels of photographic realism and 2 levels of interactional realism. The low to high photographic realism differed in terms of resolution of textures and number of triangles (both ranging from low to high respectively). Behavioral realism included no movement and no eye contact, medium behavioral realism included movement and eye contact and high behavioral realism included eye contact and movement. Low interactional realism was a speechless character and high interactional realism character included a character which addressed participants with a meaningful question. The procedure of generating models with different realism levels and their assessment are described in detail elsewhere [5]. With the aim of avoiding habituation to repetitively presented identical stimuli, we created 10 variants of the virtual character models for each of 18 realism combinations differing in terms of secondary features such as facial features and clothing. As a result, we obtained 180 unique characters that were presented to each of the participants.

4 PROCEDURE

After arriving at the laboratory, participants were asked to sign informed consent and proceeded to the adjustment of the VR helmet. Next, participants entered the VR environment and read the instructions. After that, they were asked to remain still for 5 minutes to let the physiological signals stabilize. Consequently, the calibration of the eye-tracker was conducted and if successful, the experiment started. Participants viewed 180 virtual characters in a free-viewing procedure in the Unity game engine [2]. The free-viewing procedure lasted around 50 minutes in total and was divided into 2 blocks each consisting of 90 virtual characters. The order of presentations was random. Each presentation (6 s) was preceded by a centrally located fixation point (9–12 s, randomly). Throughout the task, participants’ pupillary reactions were recorded.

5 EQUIPMENT

The stimuli was presented using Unity 2020.2.1f1 and HTC Vive Pro with disabled ability to freely rotate the view. To record the pupil data, Pupil Labs add-on was used with 200 Hz sampling frequency.

6 DATA PREPARATION

Pupil size was preprocessed with usage of the MATLAB toolbox for pupillary artifacts’ correction with recommended settings [3].

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Next, pupil data were normalized trial by trial via subtracting pre-stimulus baseline pupil size (1 s before the stimulus onset) from each following data point. The pupil diameter change in each trial was calculated as the average of the normalized pupil diameter in a time window of 2 to 4 s from the image onset. The trials with percentage of the missing data over 20 percent were excluded from the analyses which resulted in excluding 42.2 percent of the data. The data was aggregated by participant and condition.

7 ANALYSES

We conducted a rm-ANOVA analysis aimed at investigating the relationship between different types of realism and pupil diameter change. In this analysis the mean pupil diameter change in time window of 2-4s after stimulus onset was treated as the dependent variable and the three types of realisms as repeated measures factors: photographic realism (3 levels), behavioral realism (3 levels), and interactional realism (2 levels). Where sphericity assumption was not met, the Greenhouse-Geisser correction was applied.

8 RESULTS

Pupil diameter change differed significantly between photographic ($F(2, 60) = 8.879, p < .001$) and interactional realism levels ($F(1, 30) = 6.358, p = .017$). There were no significant differences across behavioral realism levels ($F(2, 60) = .250, p = .780$). High interactional realism led to significantly larger pupil diameter change than the low one. In the case of photographic realism, the differences between low and medium ($p = .002$) as well as low and high ($p = .010$) photographic realism were significant. There was no significant difference between medium and high photographic realism. None of the interactions was significant.

9 DISCUSSION

Firstly, we observed a significant effect of photographic realism on pupil diameter change with differences between low and medium as well as low and high photographic realism levels. Our exploratory study may point to the fact that people react to the manipulation of appearance physiologically. The only significant differences were between low-poly early 2000s style photographic realism and the rest. There is the possibility that this might have created a feeling of eeriness which was reflected in the participants' physiological reactions. This supposition need further studies to confirm it. Secondly, there was no effect of behavioral realism on pupil diameter change. This replicates the findings of Syrjamäki et al. [7] who observed that when a virtual companion looks at the participants it does not lead to significant change in arousal (measured by the phasic EDA) in comparison to non-looking virtual character. Thirdly, an effect of interactional realism on pupil diameter change. It might be that this effect is due to auditory stimuli being present while in the other condition it was not. If not that, it is possible that the interactional aspect of realism may be engaging enough to affect the arousal of the participants.

10 SUMMARY

This exploratory study shows that the subjective aspects of the user virtual experience may significantly influence the physiological measures of arousal. The method seems promising, especially because the pupillary measurements are becoming more and more widespread both in industrial and common usage of VR technology.

11 LIMITATIONS

One of the biggest limitations of this study is the high percentage of excluded data. Even though each realism combination was presented 10 times to each of the participants, allowing us to record more data points for each condition, it is still a flaw of this study as it might suggest that the quality of the collected data is compromised.

Probably, the data loss is due to the combination of the misfitting headset and blinks of the participants.

Moreover, bearing in mind the fact that when creating photographic realism levels we manipulated the shaders, resolution of textures etc. this might have created the differences in the brightness of the stimuli and consequently affect the pupil diameter change.

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